

AF208 User Manual



Statement

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Safety Notice

- Read the user manual carefully before setting up the Giada product.
- Disconnect the power cord before installing the internal components
- Most electronic components are sensitive to static electrical charge, please wear a wrist-grounding strap when installing the internal components.
- Don't disconnect the power cord when the system is running to avoid damage to the sensitive components caused by instantaneous surge voltage.

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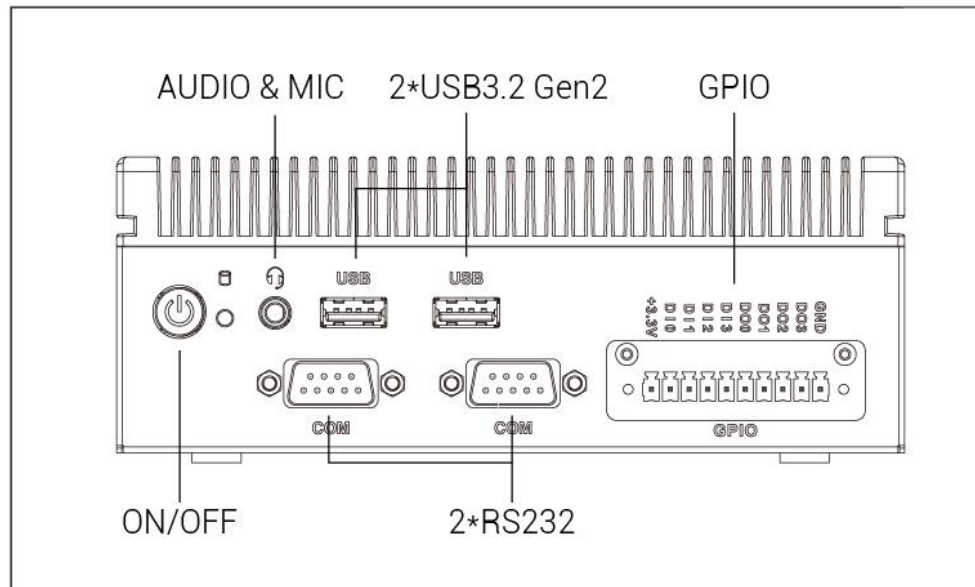
1. Product Introduction

Based on Intel® Alder Lake-N, Giada AF208 is a rugged and fanless model featuring up to 16GB onboard DDR5-4800MHz memory. Equipped with HDMI 2.0 and DP1.2 display outputs, the player delivers stunning 4K visual quality. Built to withstand demanding environments, this semi-industrial PC supports a wide temperature range (-20°C to 60°C) and voltage range (12V to 24V), making it ideal for outdoor and industrial applications. Additionally, it offers versatile connectivity with 2 x USB3.2 Gen2 ports, 2 x USB2.0 ports, 4 COM ports, and a 10-pin GPIO port, ensuring seamless integration into external devices.

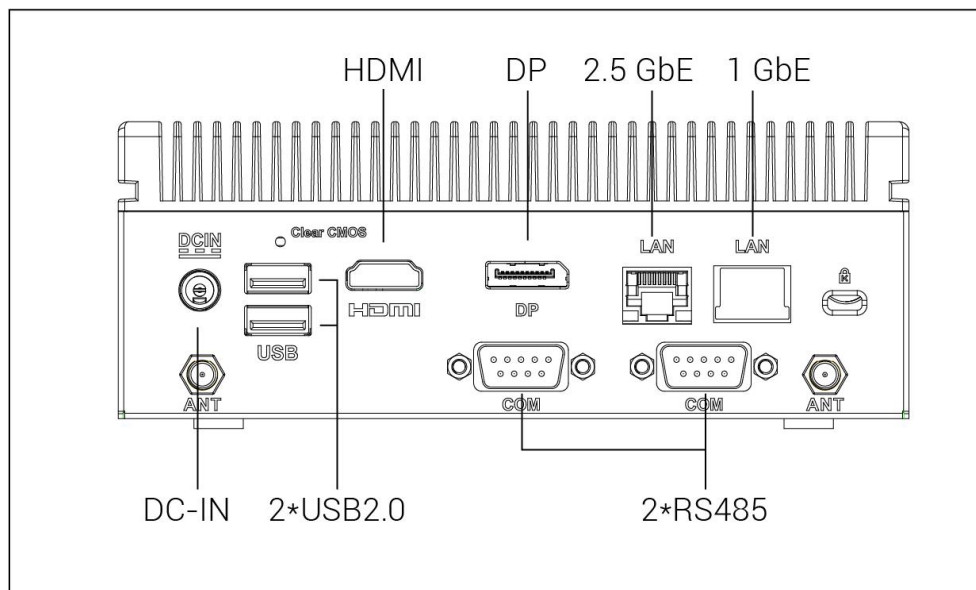
2. Interface Description and Hardware Specifications

2.1 Interface Description

Front I/O Port



Rear I/O Port



2.2 Hardware Specifications

AF208		
Processor	CPU	Intel® Processor N97 Intel® Atom X7211E Processor
	Frequency	3.60 GHz / 3.20 GHz
	Chipset	SoC
	Processing Units	4 Cores / 2 Cores
	TDP	12 W / 6 W
Memory	Type	Onboard LPDDR5-4800 MHz, Single-channel
	Max Capacity	Up to 16 GB
Graphics	GPU	Intel® UHD Graphics
	Graphic Engine	DirectX 12.1, OpenGL 4.6, OpenCL 3.0, 8K 60fps 12b 4:2:0 HEVC/VP9/SCC
	Display Interface	1 x HDMI 2.0(Max.4096 x 2160@60Hz) 1 x DP1.2(Max.4096 x 2160@60Hz) colay VGA
Network	Controller	1 x Intel i226-V Gigabit Ethernet 1 x Intel i210-AT Gigabit Ethernet
	Interface	1 x 2.5 GbE RJ45, 1 x 1 GbE RJ45
	Wi-Fi/BT	1 x E-Key M.2 (2230) for Wi-Fi/BT Support Wi-Fi 5/6/6E (CNVi)
	Mobile Network	NA

I/O Interface	USB	2 x USB3.2 Gen2 (10 Gbps), 2 x USB2.0
	Serial Port	2 x RS232, 2 x RS485
	Audio	1 x 2-in-1 headset (MIC-IN & AUDIO-OUT)
	Button	1 x Power on, 1 x CLR-CMOS
	GPIO	10pin GPIO (4 x GPI, 4 x GPO)
	Antenna	2 x Connector for Wi-Fi/BT
Storage	M.2 (2280)	1 x M.2(2280) PCIe3.0 X4 for SSD
JAHC	JAHC	Watchdog Timer / Auto Power On / RTC
Operation System	OS	Windows 10 (64bit) /Windows 11 (64bit) /Linux Ubuntu (64bit)
Power	Power Type	Lockable DC-IN
	Input Voltage	12~24 V, 45 W
Mechanical	Construction	Metal
	Dimension (W x D x H)	160 mm x 130 mm x 60 mm (6.30" x 5.12" x 2.36")
	Color	Grey
Environment	Operating Temperature	-20°C ~60°C (-4°F~140°F) @0.7m/s Air Flow
	Relative Humidity	90% @45°C (non-condensing)
Certification		CE, FCC Class B, UKCA

3. Accessories Installation Steps

▲ For safety reasons, please ensure that the power cord is disconnected before opening the case.

How to open the top cover

Unscrew the four screws and open the top case. (1 x M-Key M.2 (2280) for SSD, 1 x E-Key M.2 (2230) for Wi-Fi/BT are on top side).



3.1 WIFI (M.2) Installation

1. Unscrew the screw.
2. Plug the WIFI module into the appropriate slot.
3. Secure the module to the carrier by tightening up the screw.
4. Connect the two cables to WIFI module and install the antennas.





3.2 SSD (M.2) Installation

1. Plug the SSD (M.2) into the appropriate slot.
2. Secure the module to the carrier by tightening up the screw.



4. BIOS Setup

Notice:

The descriptions relating to BIOS setup in this Manual is for reference only when the BIOS version of the product might be upgraded. Giada provides no guarantee that all the contents in this Manual are consistent with the information you acquired.

BIOS is a basic I/O control program saved in the Flash Memory. Bridging the motherboard and the operation system, BIOS is used for managing the setup of the related parameters between them.

When the computer is activated, the system is first controlled by the BIOS program. Firstly, a self-detection called POST is performed to check all hard devices and confirm the parameters of the synchronous hardware.

Once all detections are completed, BIOS will hand over the controlling to the operation system (OS). As BIOS serves as the only channel that connects the hardware and software, whether your computer can run stably and work in optimized state will hinge on how to properly set the parameters in BIOS. Therefore, the correct setup of BIOS plays a key role for stably running the system and optimizing its performance.

The CMOS Setup will save the set parameters in the built-in CMOS SRAM on the motherboard. When the power is shut off, the lithium battery on the motherboard will provide continuously power to CMOS SRAM.

The BIOS setup program will allow you to configure the following items:

1. HD drive and peripheral devices
2. Video display type and display items
3. Password protection
4. Power management characteristics

A. State of BIOS Setup

When the computer is started up, BIOS will run the self-detection (Post) program. This program includes series of diagnosis fixed in BIOS. When this program is executed, the following information will appear if any error is found:

Press [F1] to Run General help

Press [F2] to Load previous values and continue

To enter BIOS, you can press F2 to load the default values and enter the system, you can press DEL to enter the BIOS interface if no error is found. If the indicative information disappears before you operate, you can shut off the computer and turn it on again, or you can press the RESET key on the product case. To restart your computer, you can also simultaneously press < Ctrl > + < Alt > + < Delete >.

B. Function Keys definitions

Hot Key	Description
↑	(Up key) Move to the previous item
↓	(Down key) Move to the next item
←	(Left key) Move to the left item
→	(Right key) Move to the right item
ESC	Exit the current interface
Page Up	Change the setup state, or add the values
Page Down	Change the setup state, or deduct the values
F1	Display the information of the current function Keys definitions.
F9	Load the optimized values
F10	Save the settings and exit the CMOS SETUP

C. Auxiliary information on Main interface

When the system enters the main interface of Setup, the major selected contents will be displayed at the lower part of the interface with the change of the options.

When you set the value for each column, you can view the preset value of the column and the values that can be set if you press F2, for example, the BIOS default values or CMOS Setup values.

To exit the interface for auxiliary information, press [ESC].

1) Main menu

When the system enters the CMOS Setup menu, you can see the main menu on the upper part of the screen, as shown in Figure 1.

In this main menu, you can use the left and right direction keys to select the setup items.

Once the item is selected, the lower part of the computer screen will show the details of setting.

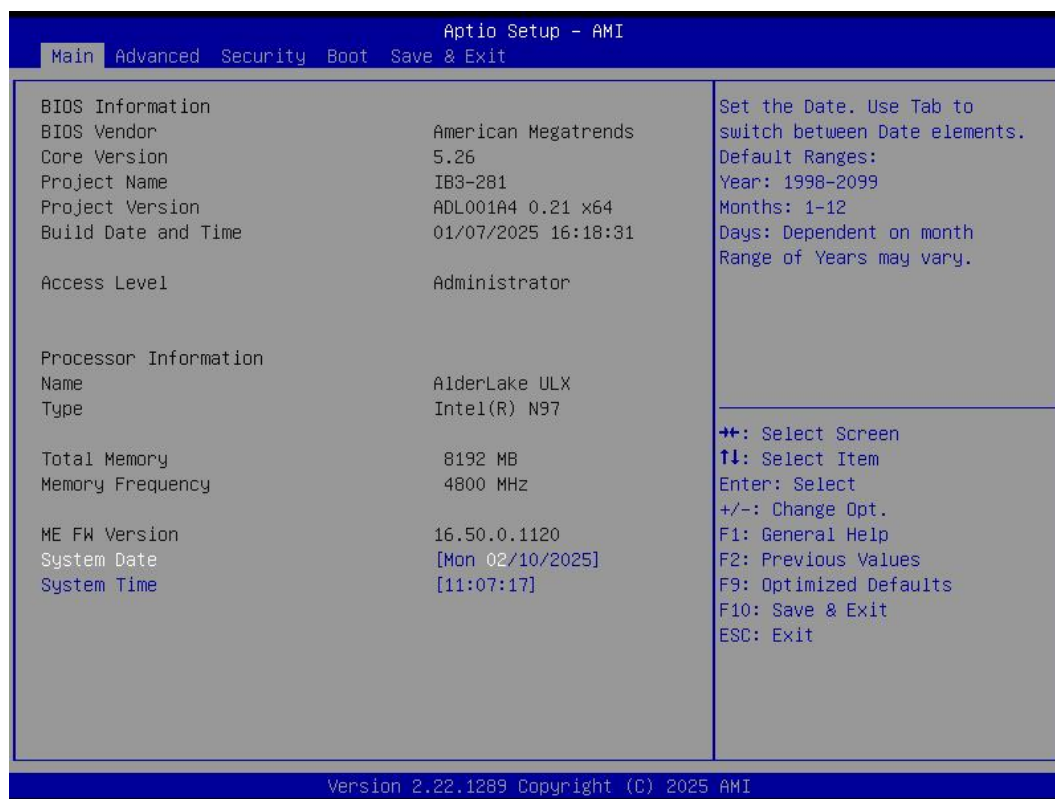


Fig 1

1) Main (standard CMOS setup)

This item is used for setting the date and time.

2) Advanced (advanced BIOS setup)

This item is used for setting the advanced functions provided by BIOS, such as specifications of PCIe facilities, CPU, HDD, etc.

3) Security (set the administrator/user password)

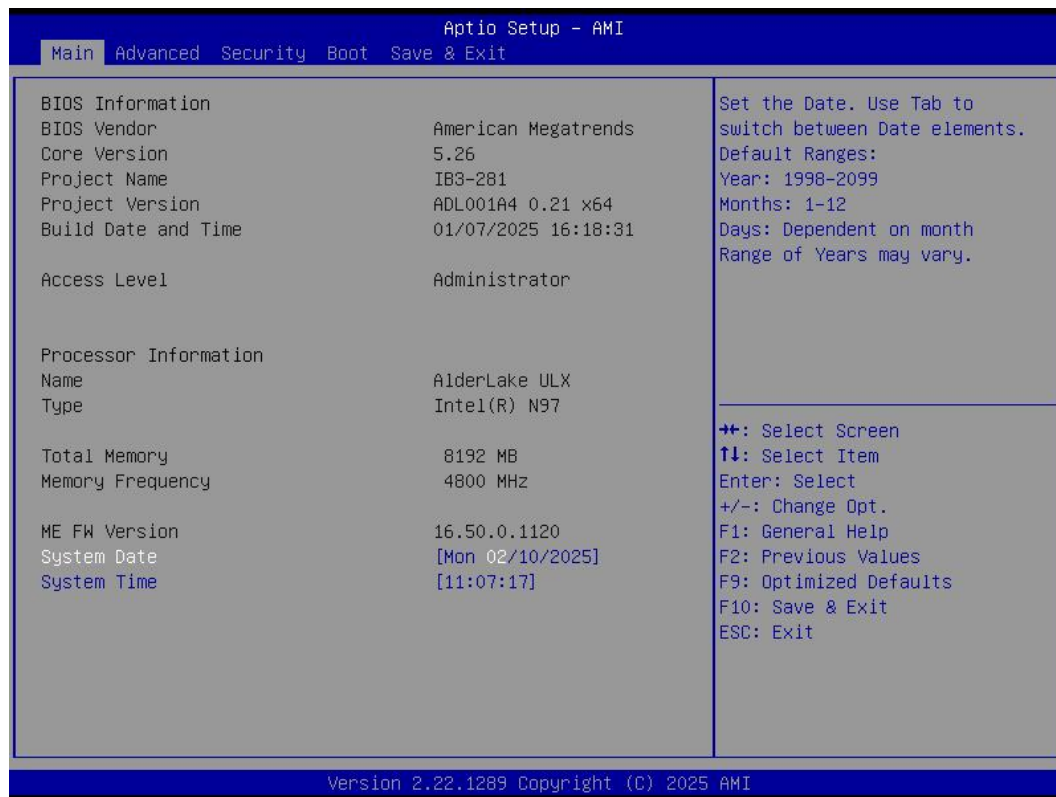
4) Boot (startup configuration characteristics)

5) Save & Exit (option of exit)

This item includes load optimal defaults / load failsafe defaults value / discard changes / discard changes and exit.

6) MEBx

4.1 Main (Standard CMOS setting)



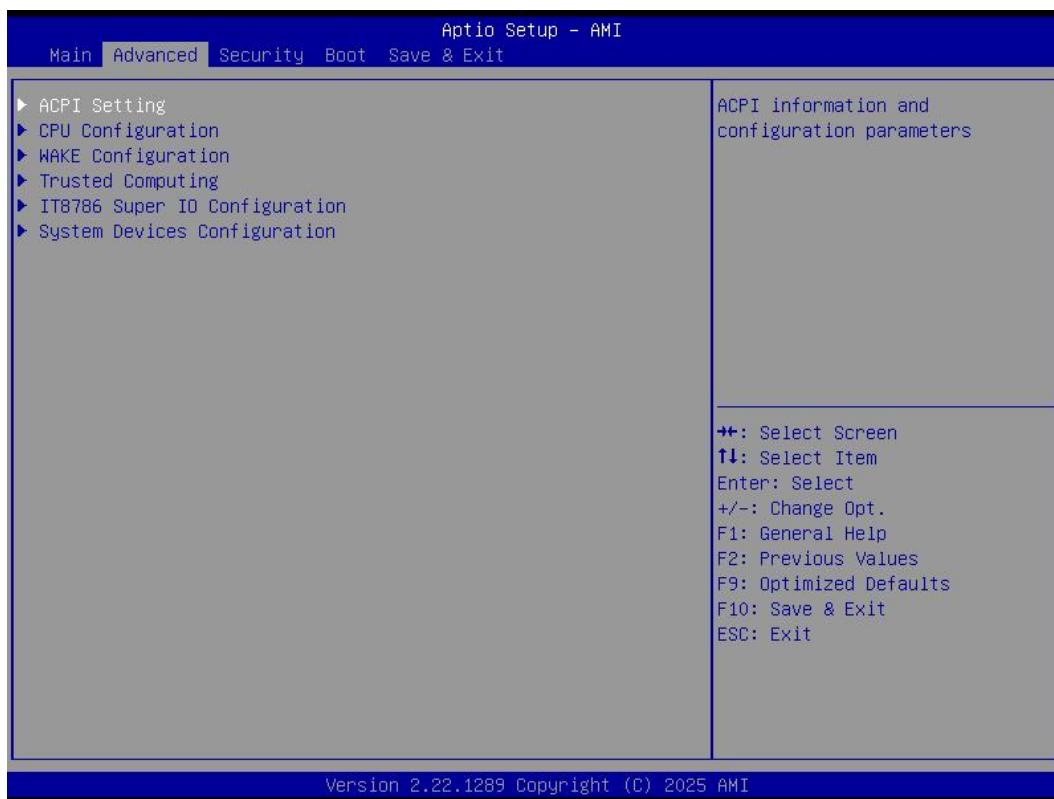
1) System time (hh:mm:ss)

Use this item to set the time for the computer, with the format as “HH / MM / SS”.

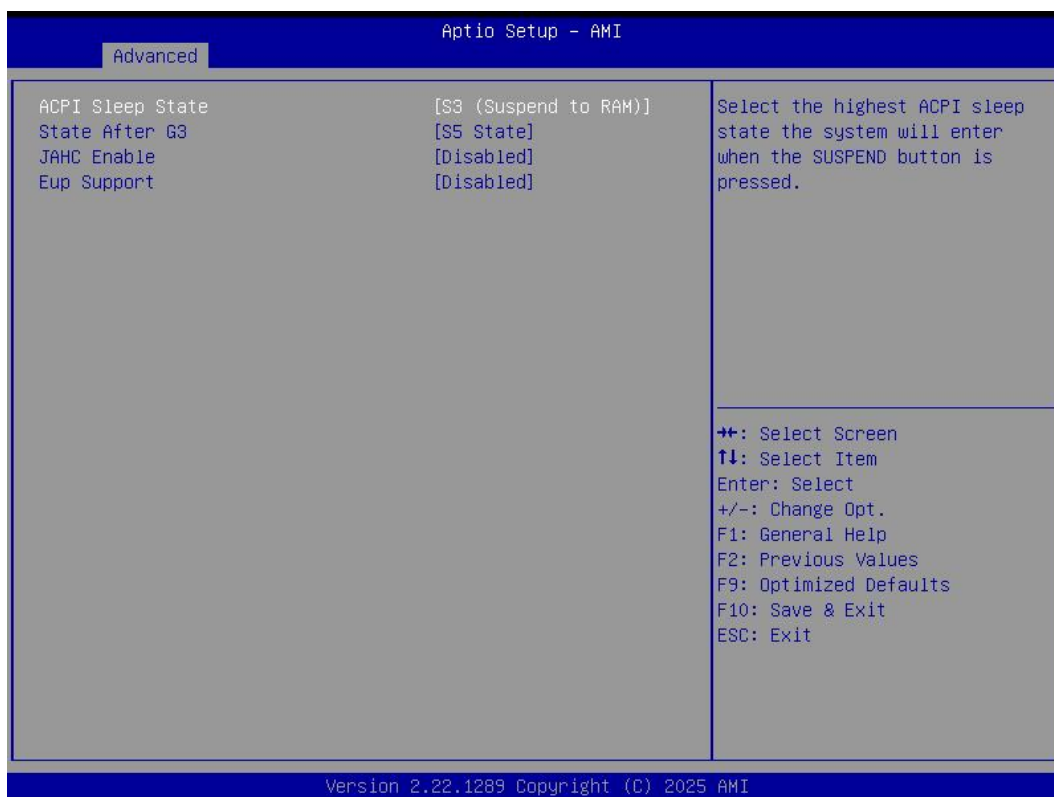
2) System date (mm:dd:yy)

Use this item to set the date for the computer, with the format as “week, MM / DD / YY”.

4.2 Advanced (Advanced BIOS setup)

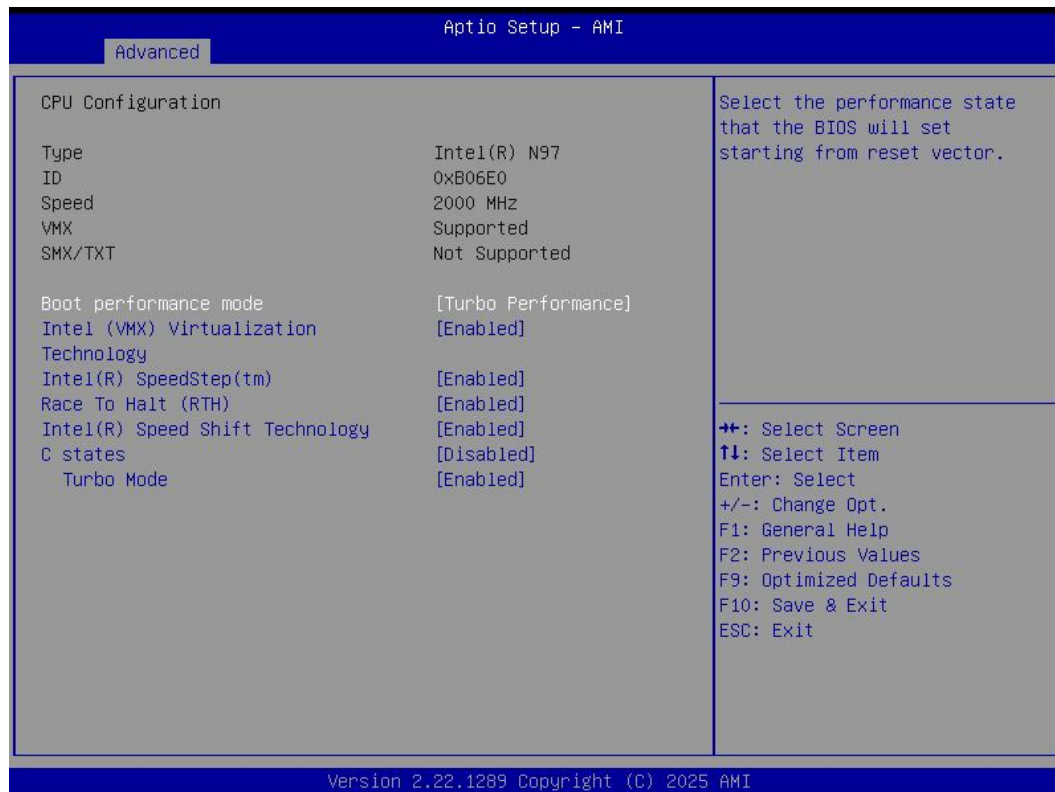
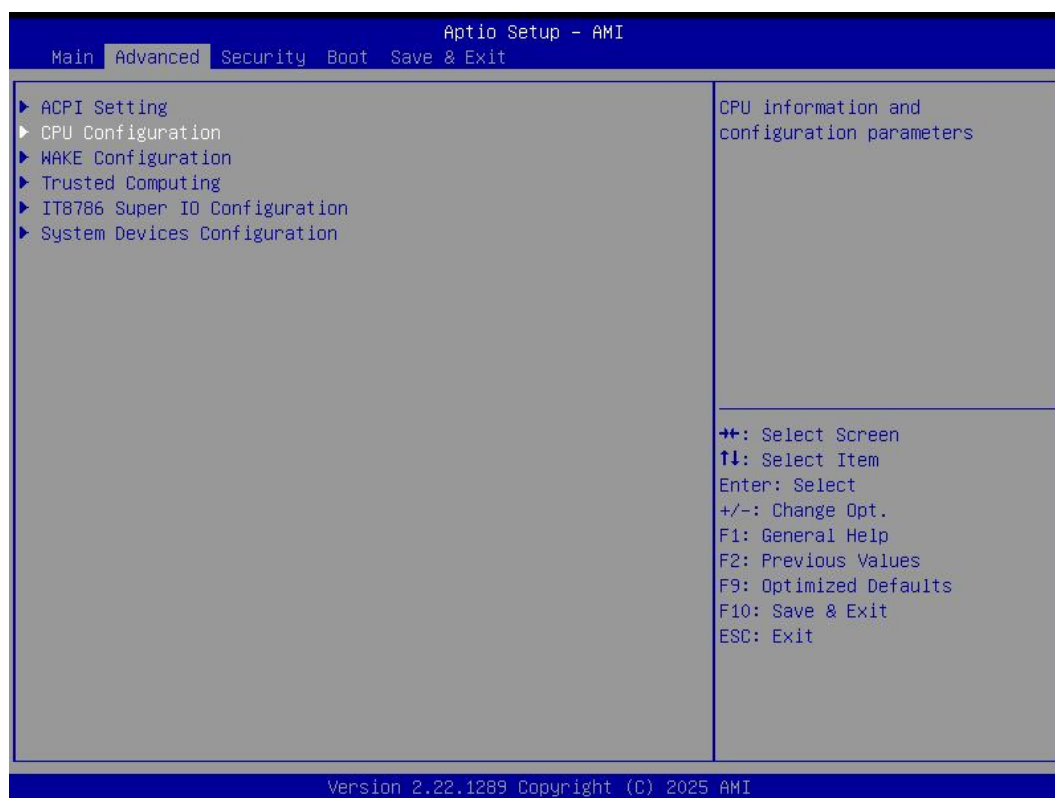


4.2.1 ACPI Setting



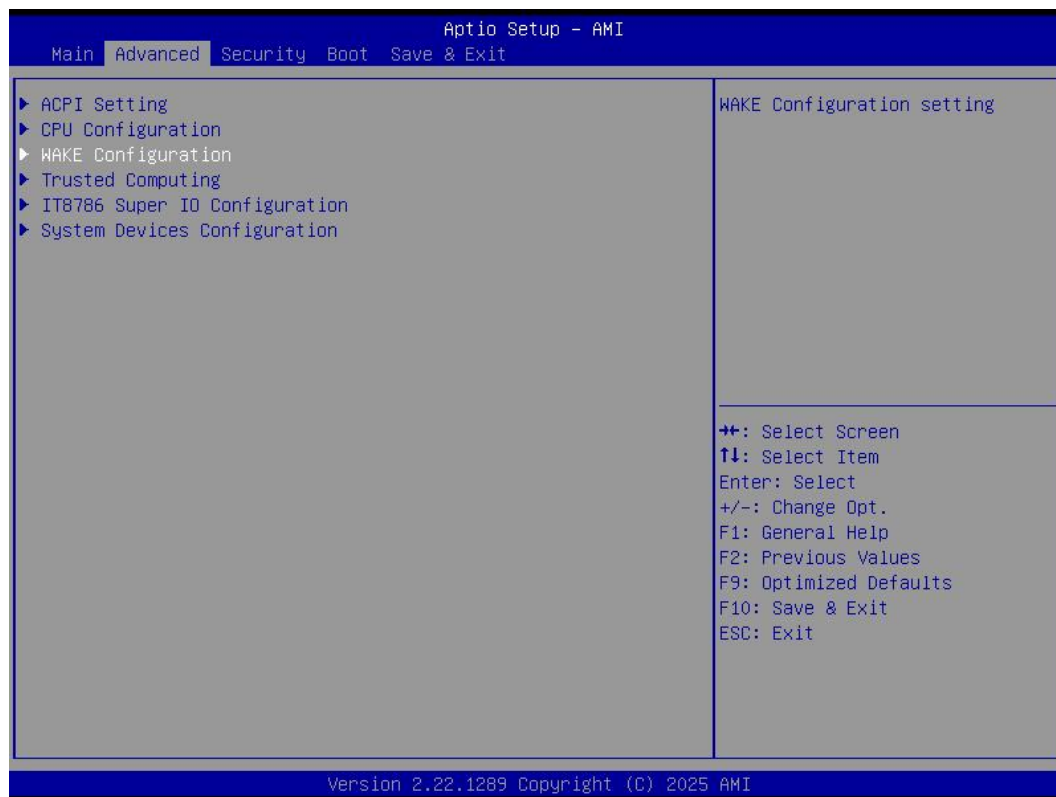
ACPI Options	Description
ACPI Sleep state	You can use the ACPI Sleep state option to control system hibernation Suspend Disabled: Disable system Suspend. ● S3 (Suspend to RAM): Enable S3(Suspend to RAM)
State After G3	● State After G3 means after restore power supply. ● S5 State (Default): If set it as S5 State, it means the system will remain shutdown state ● S0 State: If set it as S0 State, it means the system will be power on automatically.
JAHC Enable	JIEHE Active Hardware Control (JAHC) management system includes both hardware Micro Control Unit (MCU) and software (JAHC Technology Manager). ● Disabled: The JAHC is disable by default. ● Enabled.
EUP Support	EUP function: The EUP function is disable by default.

4.2.2 CPU Configuration



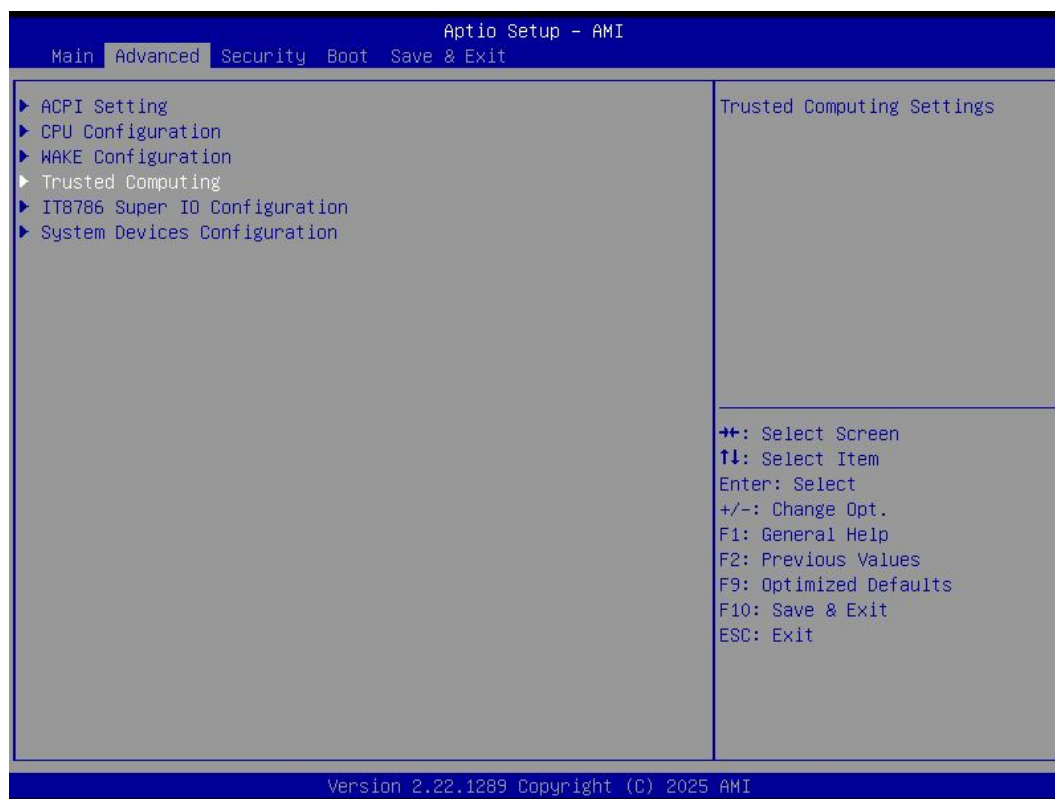
The menu	Description
CPU Configuration	
Boot performance mode	● Max Non-Turbo Performance: the best performance. ● Max Battery. ● Turbo performance.
Intel (VMX) Virtualization Technology	Intel Virtualization Technology is enabled by default. User can enable and disable the Intel Virtualization Technology function.
Intel (R) Speed Step (tm)	Intel (R) Speed Step Technology dynamically increases the processor's frequency as needed by taking advantage of thermal and power headroom to give you a burst of speed when you need it, or increased energy efficiency. The option is enabled by default. You can disable the function if it's necessary.
Race To Halt (RTH)	The Race To Halt (RTH) function is enable by default. It can adjust the CPU base frequency work in C-state. Optional: C-state.
Intel (R) Speed Shift Technology	Intel speed shift function is enabled by default. Intel® Speed Shift Technology uses hardware-controlled P-states to deliver dramatically quicker responsiveness with single-threaded, transient (short duration) workloads, such as web browsing, by allowing the processor to more quickly select its best operating frequency and voltage for optimal performance and power efficiency.
C states	The C-State function is disabled by default.
Turbo Mode	● Disabled. ● Enabled.

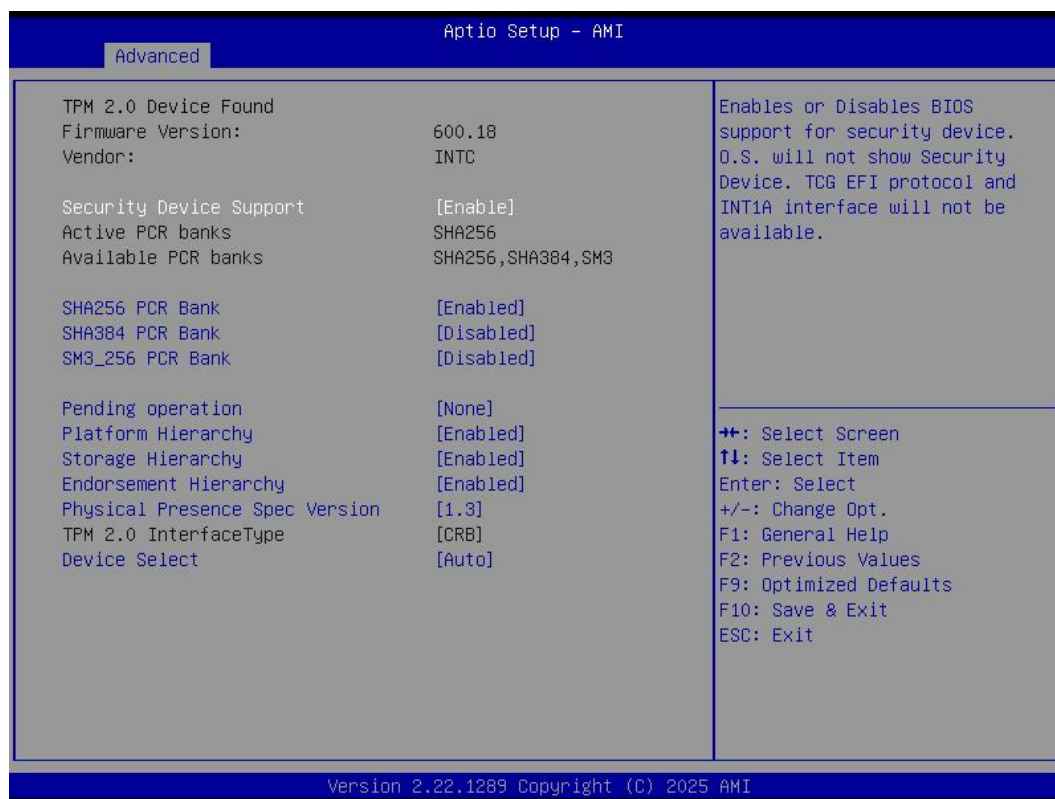
4.2.3 Wake Configuration



WAKE Configuration Options	Description
WAKE Configuration	
Wake Up On RLT LAN	Wake On LAN Function. ● Disabled: The WOL is disabled by default. ● Enabled.
Wake up By USB KB/MS	● Enabled/Disabled Wake Up by USB KB/Mouse from S3 Status.
Wake system from S5	● Disabled: By default, the functions is disabled. ● Fixed Time: You can use the fixed Time menu to modify the wake time. ● Dynamic Time: You can set multithreaded to wake system from S5 after enabling Dynamic Time.

4.2.4 Trusted Computing

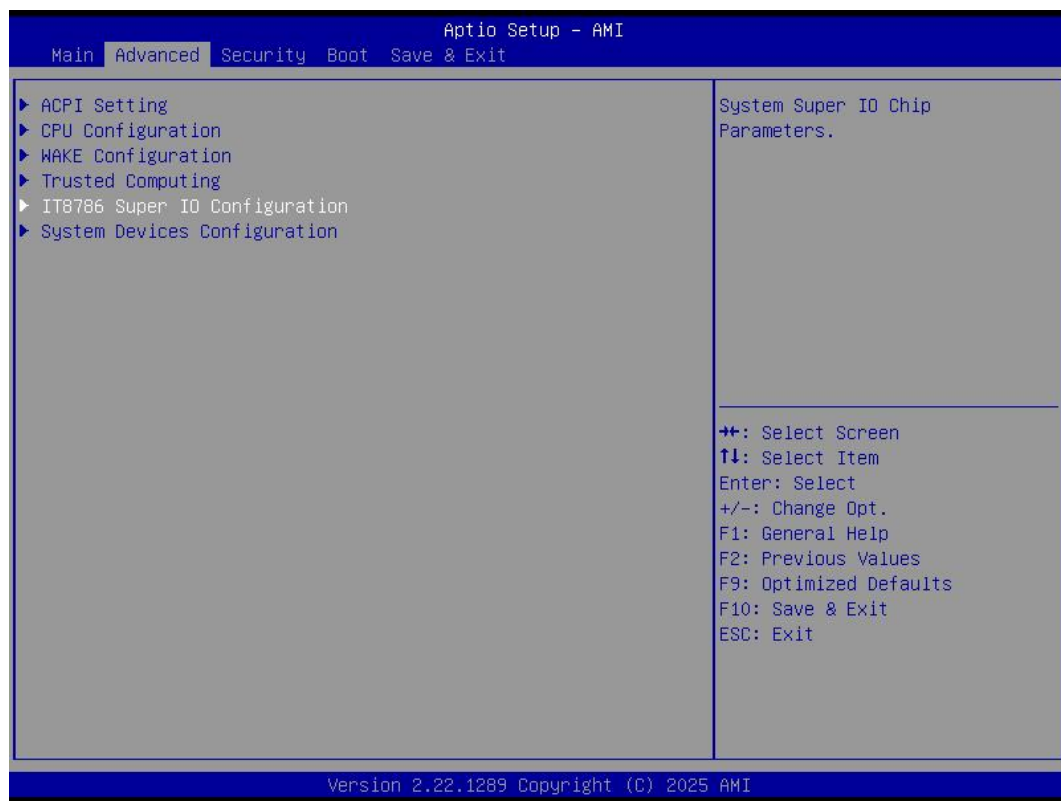




TPM20 Device Found	Description
Firmware Version	TPM FW version is 600.18
Vendor	The vendor is INTC
Security Device Support	- Disabled - Enabled. This item is enabled by default.
SHA256 PCR Bank	- Disabled. This item is Enabled by default - Enabled.
SHA384 PCR Bank	- Disabled This item is Disabled by default. - Enabled.
SM3_256 PCR Bank	- Disabled This item is Disabled by default. - Enabled.
Pending operation	It includes None and TPM Clear function.
Platform Hierarchy	Disable or Enable the Platform Hierarchy.
Storage Hierarchy	Disable or Enable the Storage Hierarchy.
Endorsement Hierarchy	Disable or Enable the Endorsement Hierarchy.

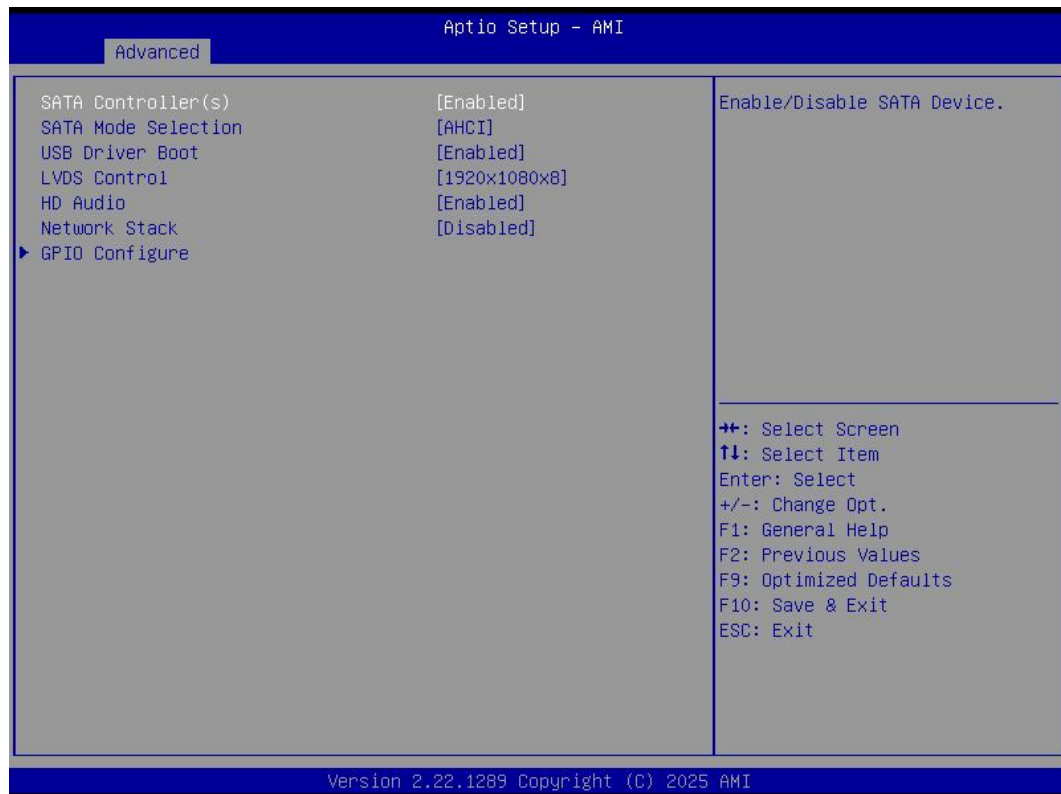
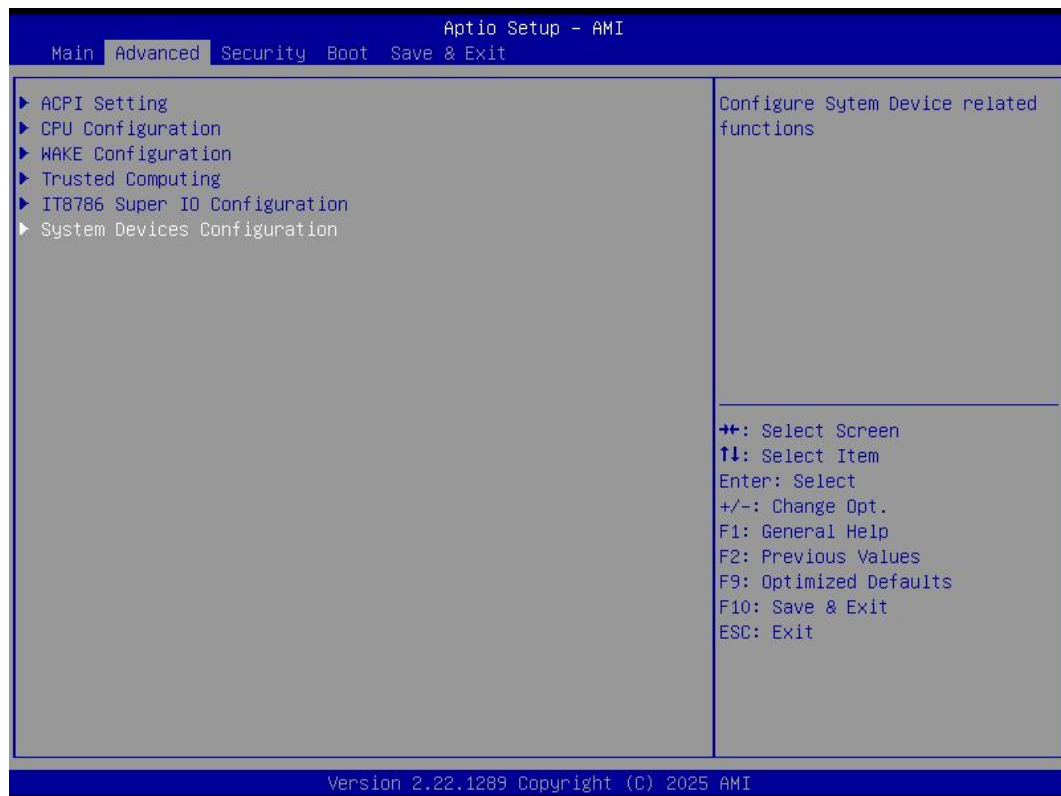
TPM20 Device Found	Description
Physical Presence Spec Version	You can choose 1.2 or 1.3. The version is 1.3 by default.
TPM 20 Interface Type	TPM2.0 Interface Type is CRB by default.
Device Select	You can select TPM1.2 or TPM2.0 or Auto. Auto is set up by default.

4.2.5 IT8786 Super IO Configuration



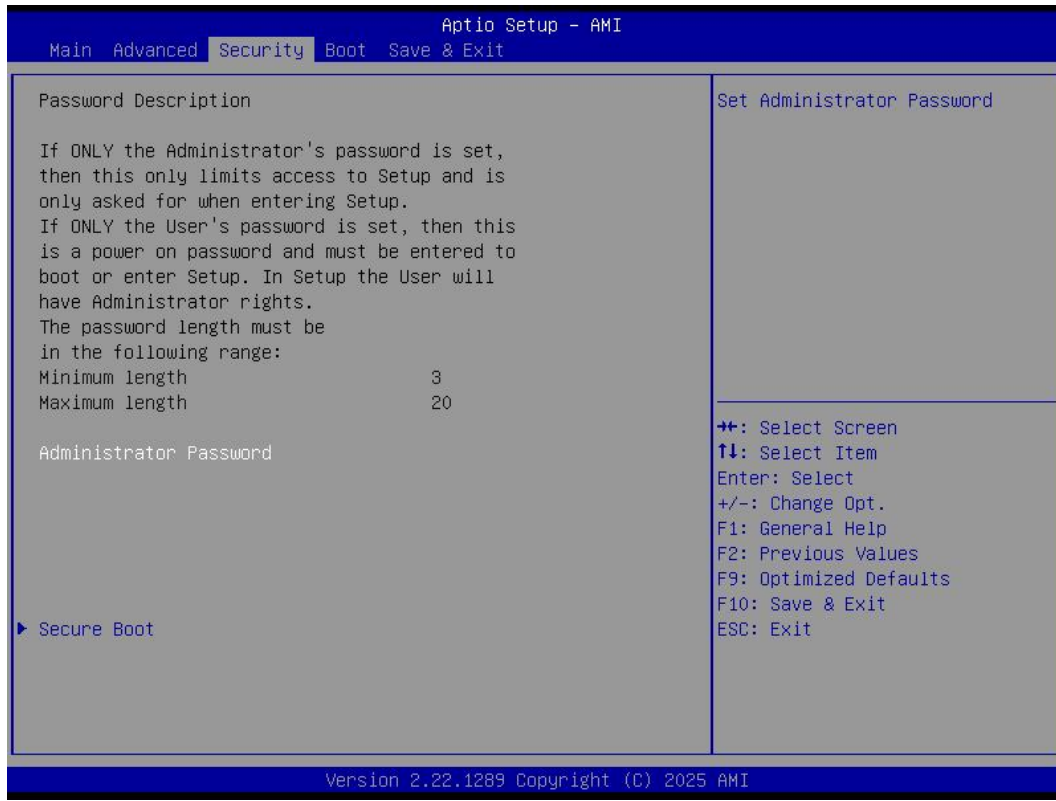
Pc Health Menu	Description
Pc Health Status is NA (AF208 is fanless design by default.)	

4.2.6 System Devices Configuration



Menu	Description
SATA Controller(s)	The SATA controller is enabled by default. ● Enabled ● Disabled.
SATA Mode Selection	● Determines how SATA controller(s) operate.
USB Driver Boot	● The USB driver Boot is enabled by default. ● Enabled ● Disabled
LVDS Control	● The LVDS option is disabled by default. ● Enabled ● Disabled
HD Audio	● Control Detection of the HD-Audio device. ● HAD will be unconditionally disabled when disabled. ● HAD will be unconditionally Enabled when Enabled.
Network Stack	The Network Stack is disabled by default. ● Enabled ● Disabled.
GPIO Configure	The GPIO port can be set High or Low.

4.3 Security Setup (set the administrator/user password)



If this function is selected, the following information will appear:

Enter New Password hhhhhh

Then enter a password which is no more than eight characters and press <Enter>. BIOS will require to enter the password again.

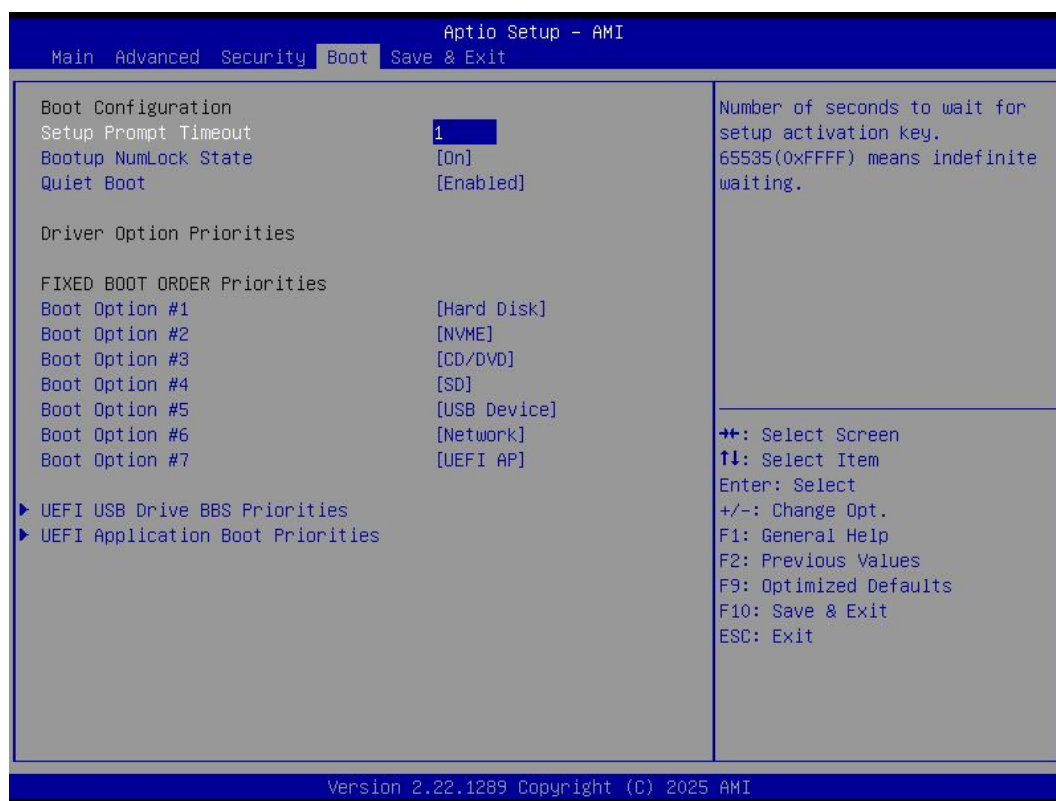
Once you enter it again, BIOS will save the set password. Once the password item is enabled, you will be required to enter the password every time before the system entering to the setup program of BIOS. The user can set this item through the Security Option in advanced BIOS properties. If the Security Option is set as System, the password will be required to be entered before both the system guides and entering to the setup program of BIOS. If it is set as Setup, the password will be required to be entered only before the system entering to the setup program of BIOS.

To delete the password, press <Enter> in the popped-up window that requires to enter the password. Then information for confirmation will appear on the screen to allow you decide whether the password will be disabled. Once the password is disabled, you can enter the setup program directly without password when the system is restarted.

Boot Sector Virus Protection. This item is used for setting the alarm function in case of virus attack in IDE disk sector. If this item is set as Enable and some program writes information in the sector, BIOS will display alarm information on the screen and buzz.

Security Item	Description
System Mode Configuration	
Secure Boot	Secure Boot feature is Active if Secure Boot is Enabled. Platform key (PK) is enrolled and the System is in user mode. The mode change requires platform reset.
Secure Boot Mode	Secure Boot Mode options includes Standard and Custom mode. In Custom mode, Secure Boot Policy variables can be configured by a physically present user without full authentication.

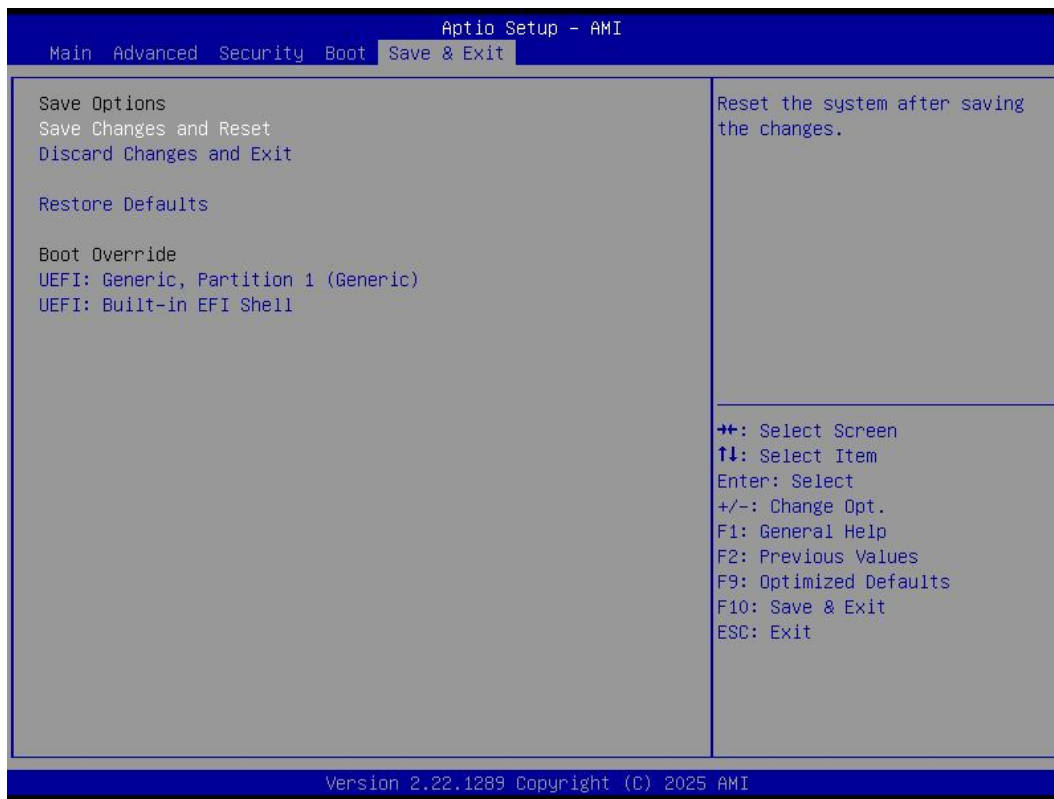
4.4. Boot Menu



Boot Item	Description
Boot Configuration	
Setup Prompt Timeout	This item is used to set the waiting time of entering the operation system. During the BIOS post, if user doesn't press the keyboard, it won't respond unless you reboot the BIOS. The Setup Prompt Timeout is 3s by default. You can set the time as you want.

Boot Item	Description
Boot up Num Lock State	Options are OFF and ON. In other words, this item can be used to set the state of Num Lock after entering the system. It can be set according to user's needs and doesn't affect the performance of the computer.
Quiet Boot	If this item is set as Enabled, the system can be started within five seconds and some detection items will be ignored. The options are [Disabled] and [Enabled].
FIXED BOOT ORDER Priorities	
Boot Option #1	The first boot device. If BIOS doesn't detect the first boot device, it will check the second boot device.
Boot Option #2	The second boot device.
Boot Option #3	The third boot device.
Boot Option #4	The fourth boot device.
Boot Option #5	The fifth boot device.
Boot Option #6	The sixth boot device.
Boot Option #7	The seventh boot device.
UEFI USB Drive BBS Priorities	You can set and management UEFI USB device after enabling this option.
UEFI Application Boot Priorities	You can set and management UEFI hard disk after enabling this option.

4.5 Save&Exit



Save Exit Item	Description
Save Options	
Save Changes and Reset	Save all changes and exit
Discard Changes and Reset	Give up the settings and exit.
Restore Defaults	Recover it to default.
Boot Override	Whole Boot devices

5. JAHC Introduction

JEHE Active Hardware Control (JAHC) management system includes both hardware Micro Control Unit (MCU) and software (JAHC Technology Manager). It can support following functions:

1. Automatically boot up when power on. It is controlled by the Micro Control Unit (MCU) chip.
2. Real Timer Controller (RTC) wake up: user can install the JAHC software to set up automatic startup and shutdown, one week as a circle.
3. Watchdog timer. It is a built-in API interface.
4. Infrared remote control (Optional IR controller).

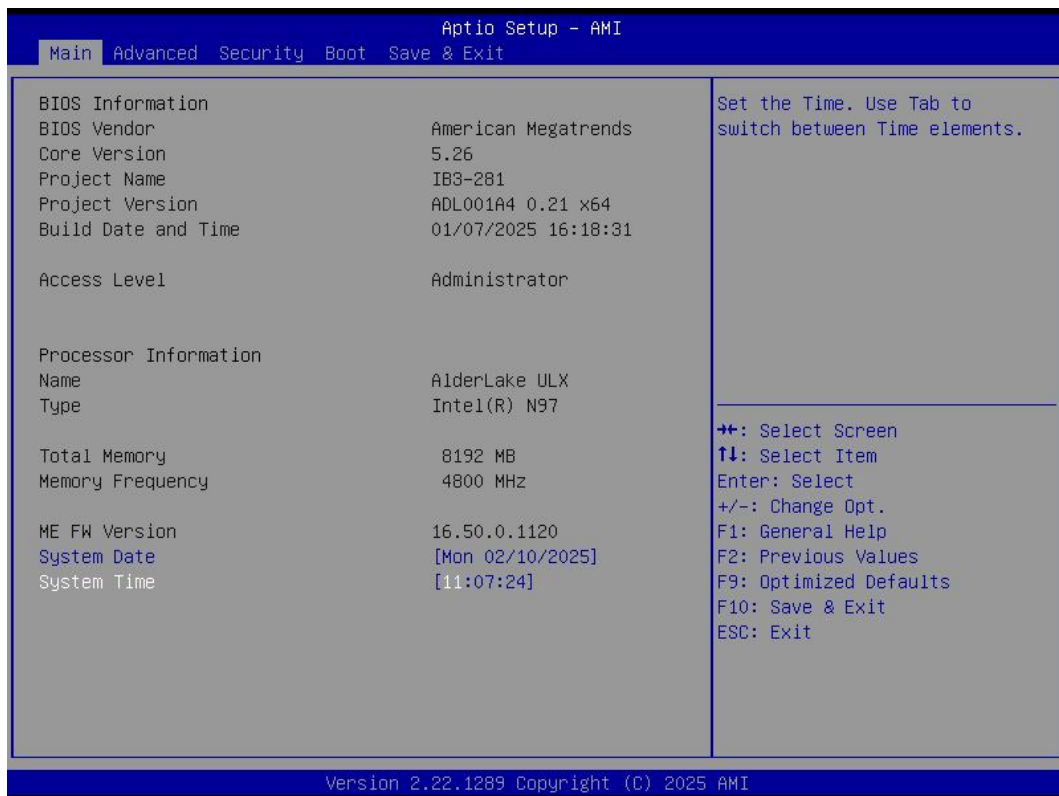
5.1 How to set up Auto power on function

Automatically reboot when power on

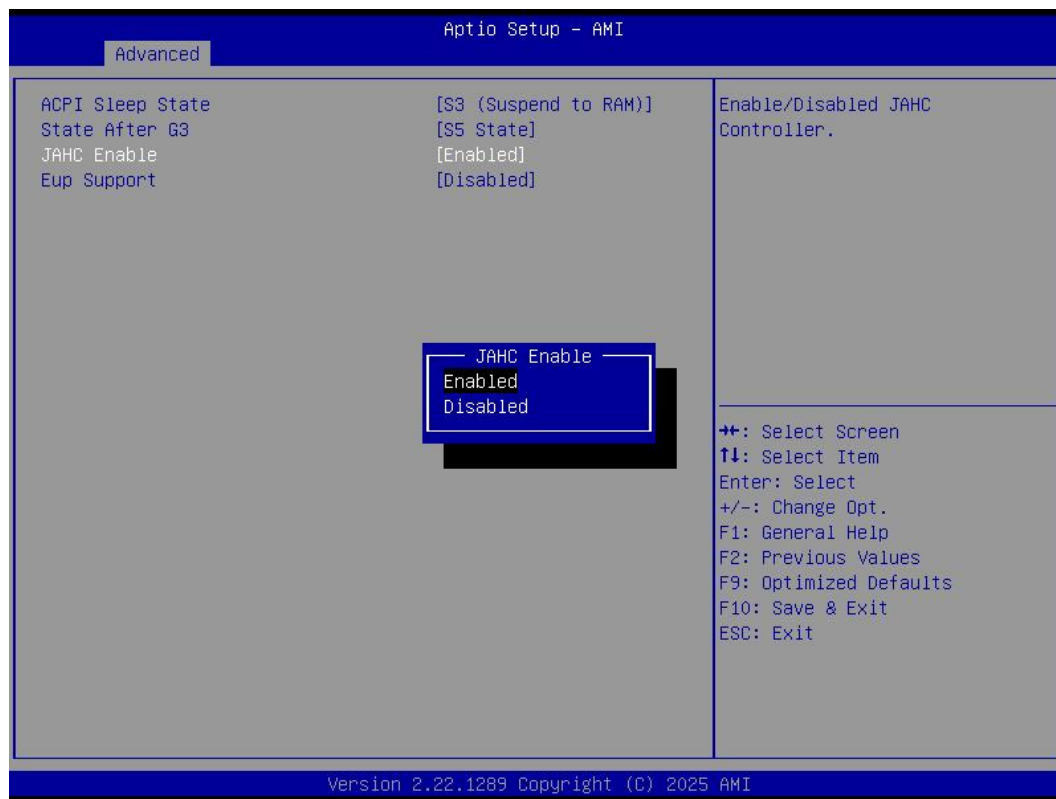
The function of automatically reboot when power on is controlled by hardware. You can enable it by switching the JAHC button to “on”.

If you cannot find the physical switch on the player, then you can go into the BIOS to enable it by following steps:

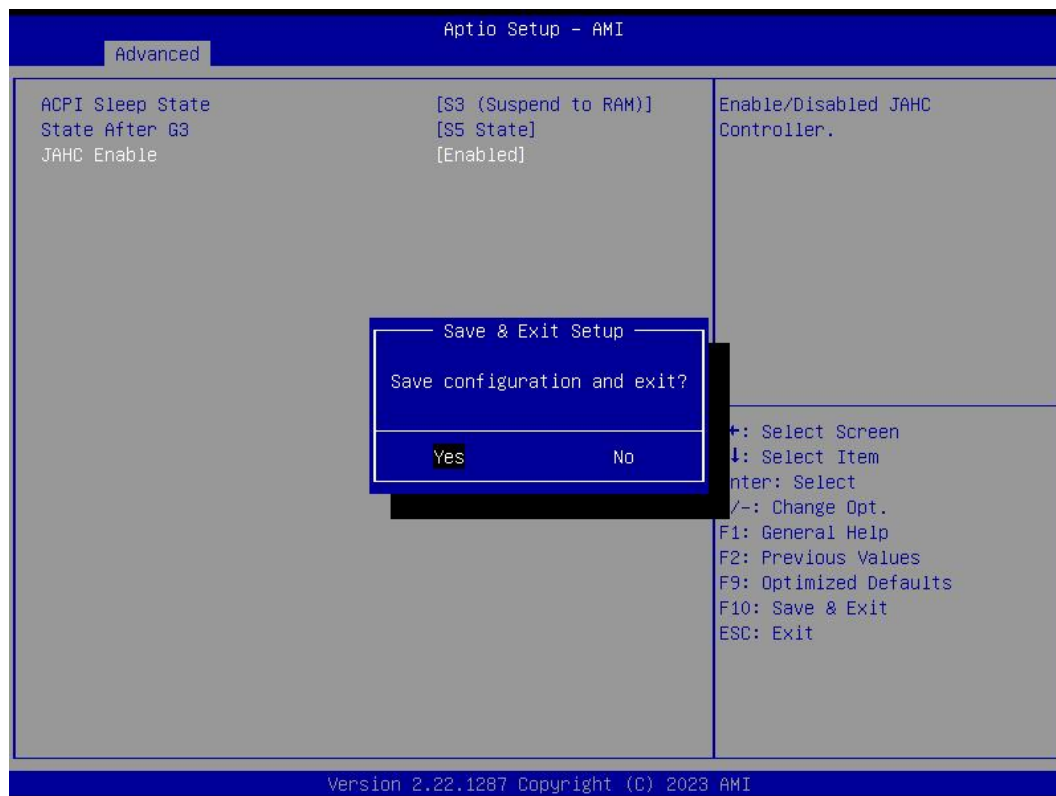
- a. Turn on the player and continually press ‘Del’, then it can enter BIOS setup menu.



- b. Select Advanced- > ACPI Setting- > JAHC Enable- > Enabled.



c. Press 'F10' to save change & exit after select "JAHC enabled" option.



5.2 JAHC software

5.2.1 JAHC software functions

- RTC wake up. The user can set up automatic startup and shutdown, one week as a circle
- Caution message prior to shutdown to remind user to save the data. User can also choose to postpone the shutdown process.
- When JAHC is running, it can support reboot automatically when system is crashed. No additional settings needed.

5.2.2 JAHC software installation guide

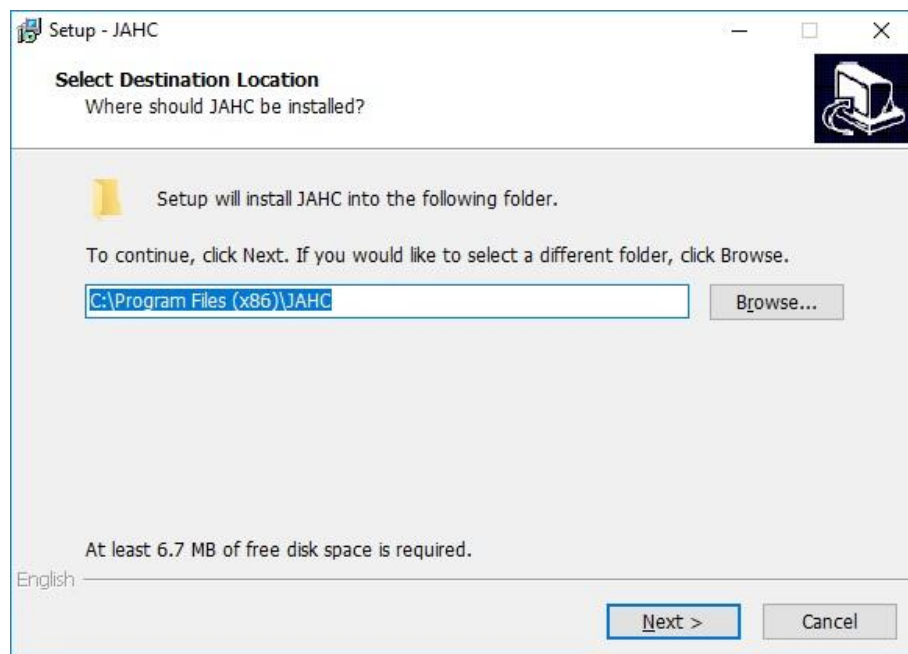
System Requirements:

- Giada player with JAHC function.
- Switch the JAHC button to “on” or enable it in BIOS if there is no physical button on the chassis.
- Supported operation system: Windows 10 64bit, Linux 64bit.

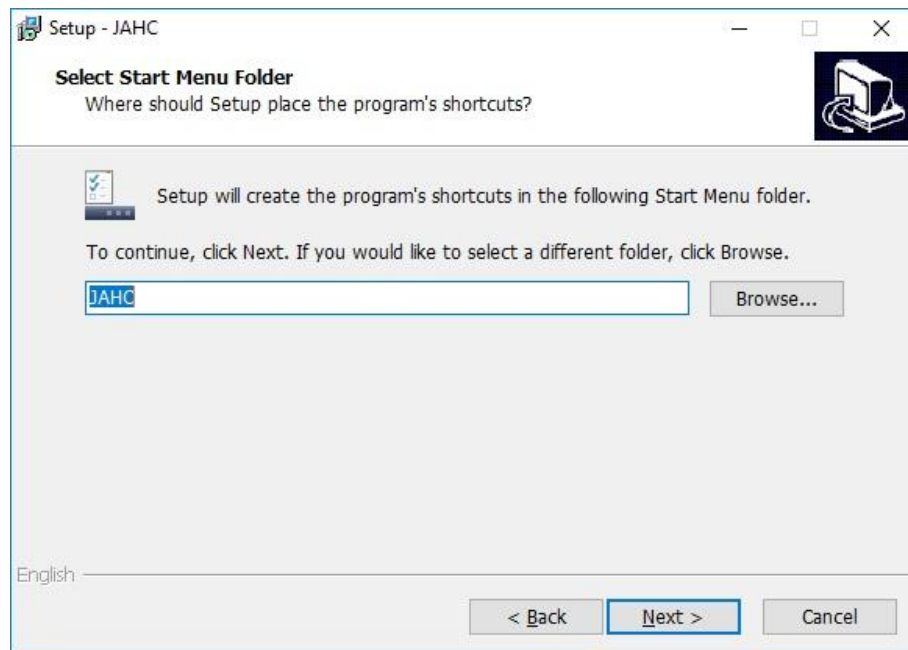
How to install JAHC software:

Please download the JAHC.EXE from Giada DVD, then follow up below steps:

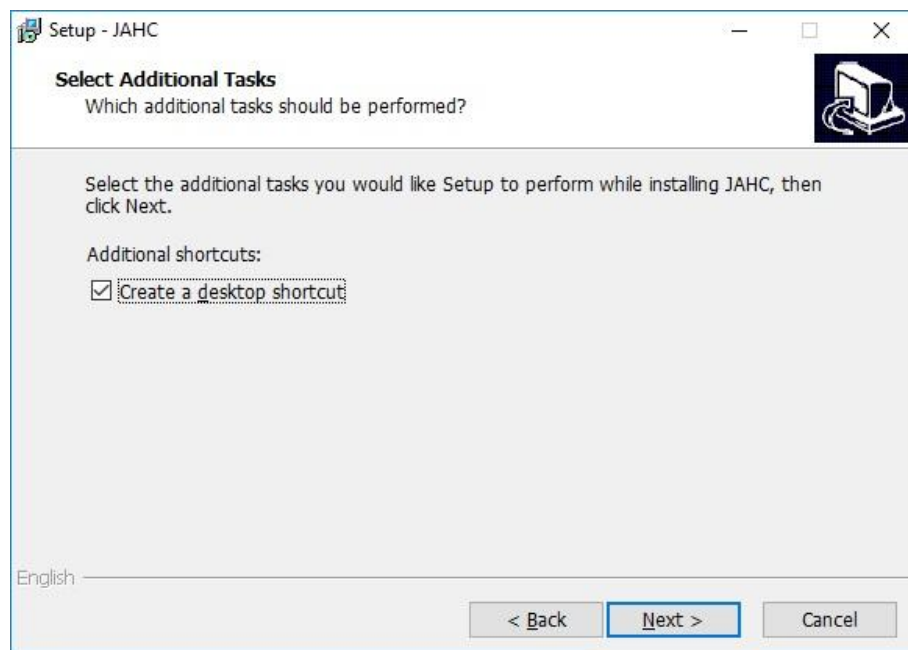
- Double-click the JAHC.EXE file, the setup wizard will pop up, select destination location and click [Next] button to continue the installation.



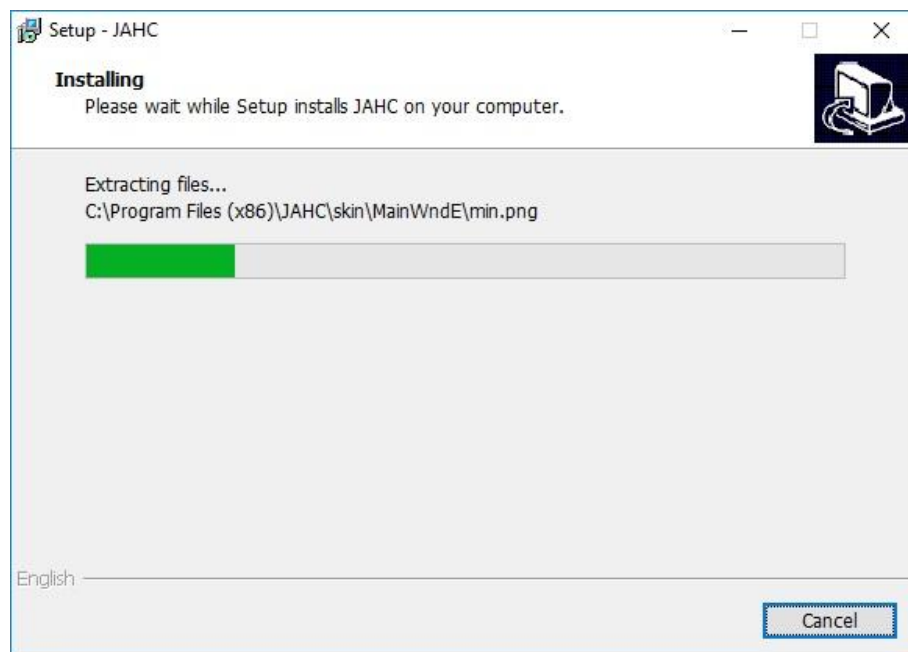
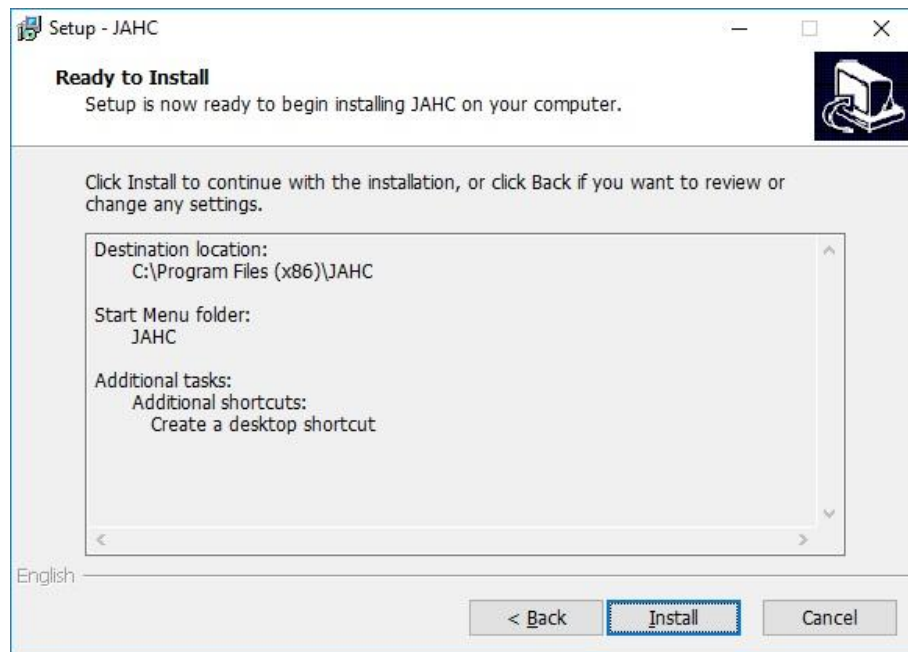
- Click [Next] button to continue the installation.



c. Select [Create a desktop shortcut] and click [Next] button.



d. Click [Install] button to continue the installation.



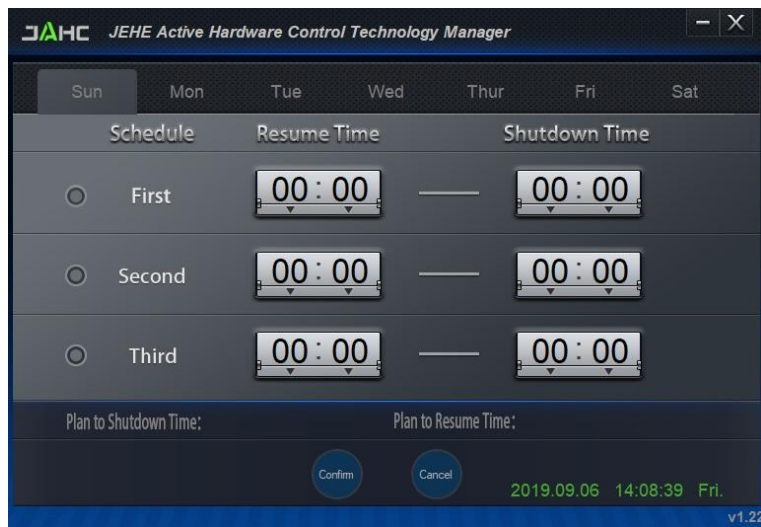
e. Click [Finish] button to finish the installation. You can select [Launch JAHC] to run the software automatically after finishing the installation.



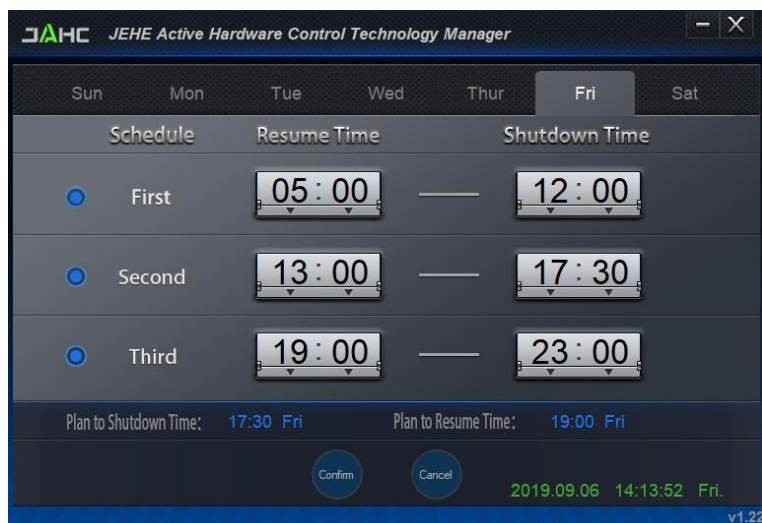
Notice: The JAHC will be added into boot item when it is installed. It will start up when system boot up.

5.2.3 Startup & shutdown time setup

After install the JAHC software, double click the JAHC icon on taskbar and the setup menu will pop up.



One week as a circle, maximum 3 schedules per day. Select each schedule to set up the resume time and shutdown time. Click [Confirm] button to launch the schedule.



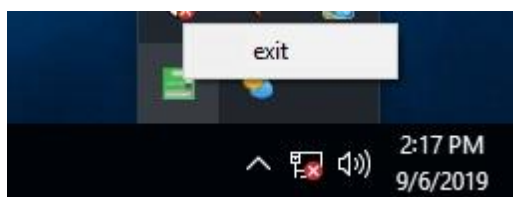
After finishing the setup, the menu window will notice the resume time and shutdown time.

⚠ Caution: If the interval from shutdown time to next resume time is less than 3 minutes, the system will not shut down.

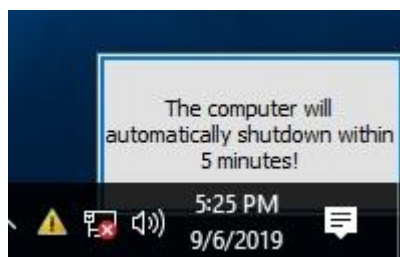
Click [Cancel] button to restore the time settings and cancel the shutdown status.

Click [X] button to hide the menu. You can find it on taskbar.

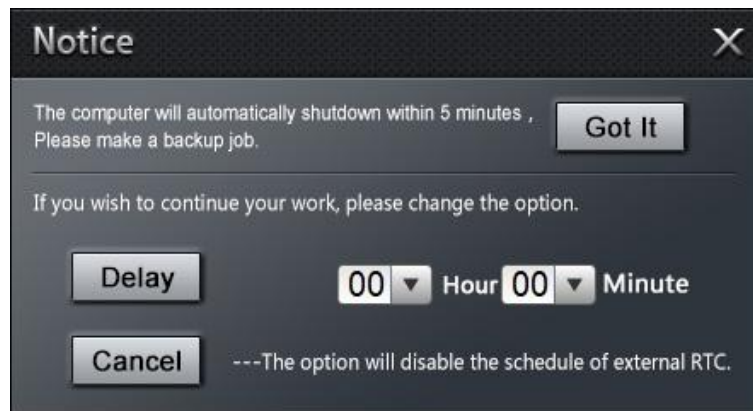
Right click the JAHC icon on taskbar and select [exit] to exit the software.



Shutdown caution: the shutdown caution will pop up before the system shutdown.



You can double click the message window and a new dialog box will pop up.



You can click [Delay] button and set up the time to delay the shutdown or click [Cancel] button to cancel the shutdown.

5.3 Watchdog API and instruction

Please contact Giada FAE (email:support@giadatech.com) for watchdog API software and instruction.



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